

# FleetSpark — User Manual

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**Audience:** Fleet managers, admins, drivers using FleetSpark

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## Getting Started

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### Creating Your Account

1. Navigate to `https://fleetspark.syncsphere11c.com`
2. Click **"Sign Up"**
3. Enter your organization name, your name, email, and password
4. You'll be the **Admin** of your organization
5. You'll land on the Dashboard

### Inviting Team Members

1. Go to **Settings** → **Users**
2. Click **"Invite User"**
3. Enter their email and select a role:
  - **Admin** — Full control, can manage users and billing
  - **Fleet Manager** — Can manage vehicles, drivers, and view all data
  - **Driver** — Can only see their assigned vehicle
  - **Viewer** — Read-only access to dashboard and reports
4. They'll receive an email invitation to set their password

## First-Time Setup Checklist

- ☐ Create your account
  - ☐ Invite team members
  - ☐ Set up your organization profile (Settings → Organization)
  - ☐ Add vehicles
  - ☐ Add drivers
  - ☐ Assign drivers to vehicles
  - ☐ Set up GPS devices (or use demo mode)
  - ☐ Configure alert thresholds (Settings → Alerts)
  - ☐ Test the dashboard
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## Dashboard

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The dashboard is your home screen. It gives you a real-time overview of your entire fleet.

### What You See

#### Summary Cards (top):

- **Total Vehicles** — All active vehicles in your fleet
- **Active Now** — Vehicles currently in transit (% of fleet)
- **Charging** — Vehicles currently plugged in
- **In Maintenance** — Vehicles scheduled for or undergoing maintenance
- **Alerts** — Unacknowledged alerts requiring attention

#### Fleet Map:

- Each vehicle appears as a colored dot
- **Green** = In Transit | **Blue** = Charging | **Gray** = Idle
- **Red** = Has an alert | **Purple** = Maintenance | **Dark** = Offline
- Click any dot to see vehicle details
- Use the legend to understand colors

#### Alert Panel:

- Shows the latest unacknowledged alerts
- Click "**View All**" to see the full alerts page
- Red = Critical, Orange = Yellow = Warning, Blue = Info

#### Vehicle List:

- Table of all vehicles with key info
- Search by name or plate number
- Filter by status (transit, charging, idle, etc.)
- Click any row to open the Vehicle Detail page

## Understanding Vehicle Status

| Status      | Icon | What It Means                         |
|-------------|------|---------------------------------------|
| In Transit  | □    | Vehicle is moving                     |
| Charging    | □    | Vehicle is plugged in and charging    |
| Idle        | ○    | Vehicle is parked, not charging       |
| Maintenance | □    | Vehicle is in the workshop            |
| Alert       | □    | Vehicle has an active alert           |
| Offline     | ●    | No GPS signal received in 30+ minutes |

## Managing Vehicles

### Adding a Vehicle

1. Click "**Vehicles**" in the sidebar
2. Click "**Add Vehicle**" button
3. Fill in the required fields:
  - **Name** — e.g., "Kayoola-001" (your internal name)
  - **Plate Number** — e.g., "UAX 123A"
  - **VIN** — Vehicle Identification Number (optional but recommended)
  - **Model** — e.g., "Kayoola EVS"
  - **Year** — e.g., 2025
  - **Type** — Select from: Bus, Coach, Car, Van, Truck
  - **Battery Capacity (kWh)** — e.g., 250
  - **Max Range (km)** — e.g., 300
  - **Depot** — Where the vehicle is based (optional)
4. Click "**Save**"

### Editing a Vehicle

1. Click on the vehicle row in the list
2. Go to the "**Settings**" tab
3. Click "**Edit**"
4. Update any field
5. Click "**Save**"

### Assigning a GPS Device

1. In Vehicle Settings, find the **Device ID** field
2. Enter the ID of your GPS device (e.g., "teltonika-abc123")
3. Make sure the device is already sending telemetry (see [GPS Device Setup](#))
4. Once telemetry is received, the vehicle will appear on the map

## Deactivating a Vehicle

1. In Vehicle Settings, toggle **"Active"** to off
2. The vehicle will be hidden from the main list but data is preserved
3. To reactivate, toggle back on

## Vehicle Detail Page

Click any vehicle to see its detail page with tabs:

### Overview Tab:

- Current location on mini-map
- Current battery level, SoH, estimated range
- Today's trips summary
- Assigned driver info
- Vehicle info (odometer, battery capacity, etc.)

### Location Tab:

- Full-size map with the vehicle's current position
- Route trail showing where it's been (last 1 hour by default)
- Change time range: Last 1h, 6h, 24h, 7 days
- Geofence status (if configured)

### Battery Tab:

- Large battery gauge showing current level
- State of Health (SoH) percentage
- Estimated remaining range
- Battery history chart (last 24h, 7 days, 30 days, 90 days)
- Charging sessions table with dates, levels, and energy consumed

### Trips Tab:

- List of all trips with date, distance, duration, avg speed
- Click a trip to see the route on a map
- Filter by date range

### Maintenance Tab:

- Upcoming maintenance (next 30 days)
- Maintenance history
- **"Schedule Maintenance"** button to add new

### Settings Tab:

- Edit vehicle info
- Configure max range (for accurate range estimates)
- Configure alert thresholds for this specific vehicle

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## Managing Bikes

Bikes (e-bodas, e-motorcycles) are managed similarly to vehicles but with bike-specific fields.

## Adding a Bike

1. Click **"Vehicles"** → **"Add Vehicle"**
2. Select **Type: Bike** (or "Motorcycle")
3. Fill in:
  - **Name** — e.g., "Spiro-001"

- **Plate Number** — e.g., "UAX 500M"
- **Model** — e.g., "Spiro Max"
- **Battery Capacity (kWh)** — e.g., 2.5 (much smaller than buses)
- **Max Range (km)** — e.g., 80

4. Click "**Save**"

## Bike-Specific Features

- **Battery swap tracking:** FleetSpark detects when a battery swap occurs ( sudden level jump from low to high)
- **Swappable battery ID:** Optionally assign a battery ID to track individual batteries
- **Smaller battery gauge:** UI adapts to show appropriate range for bikes
- **Bike on map:** Shown with a motorcycle icon (🏍️) vs bus icon (🚌)

## Managing a Bike Fleet

Bikes appear alongside vehicles in the dashboard. Use the **Type filter** to show only bikes.

For large bike fleets (100+):

- Use the **search** function to find specific bikes
- **Bulk import** via CSV (admin feature in v2)
- Assign **battery swap stations** as sub-locations

## Battery Swap Detection

FleetSpark automatically detects battery swaps:

1. Battery level drops to <10%
2. Suddenly jumps to >90% within 5 minutes (without charging status)
3. System logs a "Battery Swap" event
4. Visible in the Battery tab as a swap marker

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## Managing Drivers

### Adding a Driver

1. Click "**Drivers**" in the sidebar
2. Click "**Add Driver**"
3. Fill in:
  - **Name** — Full name
  - **Phone** — With country code (e.g., +256...)
  - **Email** — Optional
  - **License Number** — Optional
  - **License Expiry Date** — Optional
4. Click "**Save**"

### Assigning a Driver to a Vehicle

1. Open the **Driver Detail** page
2. Click "**Assign to Vehicle**"
3. Select the vehicle from the dropdown
4. Click "**Confirm**"

5. The driver will now appear on the vehicle's detail page

## Reassigning a Driver

1. Open the **Vehicle Detail** page
2. Click **"Change Driver"** in the driver section
3. Select a new driver or choose **"Unassign"**
4. Click **"Confirm"**

## Driver View

When a driver logs in:

- They only see their assigned vehicle(s)
  - They can see current battery level and estimated range
  - They can update their location manually (if no GPS device)
  - They cannot see other vehicles or fleet analytics
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## Battery Monitoring

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### Understanding Battery Level

- **Gauge:** Visual indicator from 0% to 100%
- **Color coded:** Green (>50%), Yellow (20-50%), Red (<20%)
- **Estimated Range:** Calculated from battery level × max range

### Understanding State of Health (SoH)

- **SoH** measures battery degradation over time
- **100%** = Brand new battery
- **80%** = Battery has degraded 20% (needs monitoring)
- **<70%** = Battery replacement recommended
- SoH naturally decreases with charge cycles

## Charging Sessions

A charging session is recorded when:

1. Vehicle reports `charging: true`
2. Vehicle reports `charging: false`
3. System calculates: duration, start/end level, estimated energy consumed

## Low Battery Alerts

- Default threshold: **20%**
- Configurable in **Settings → Alerts**
- Notifications: In-app, email, or SMS
- Acknowledge the alert to clear it from the active list

## Battery History Chart

- View battery level over time
  - Toggle: 24 hours, 7 days, 30 days, 90 days
  - Charging periods highlighted in blue
  - Useful for identifying battery degradation patterns
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## Trip History

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### What is a Trip?

A trip is automatically recorded when:

1. The vehicle starts moving (ignition on / speed > 0)
2. The vehicle stops for more than 5 minutes (ignition off)

### Trip Data

- **Start/End time** — When the trip began and ended
- **Start/End location** — GPS coordinates and reverse-geocoded address
- **Distance** — Kilometers covered (from GPS)
- **Duration** — Total time in minutes
- **Average speed** — Mean speed during the trip
- **Maximum speed** — Highest speed recorded

### Viewing Trip Route

1. Open a trip from the list
2. Click "**Show Route**"
3. The map displays the exact path taken
4. Colored by speed (green = fast, red = slow)

### Export Trips

1. Go to **Reports**
  2. Select **Trip Export**
  3. Choose date range
  4. Click "**Export CSV**"
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## Maintenance Scheduling

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### Creating a Maintenance Record

1. Open the **Vehicle Detail** → **Maintenance** tab
2. Click "**Schedule Maintenance**"
3. Fill in:
  - **Type:** Routine, Repair, Inspection, Battery Service, Tire, Brake, Other

- **Description:** What needs to be done
- **Scheduled Date:** When it should happen
- **Odometer:** Current reading (optional)

4. Click **"Save"**

## Completing Maintenance

1. Open the maintenance record that's due
2. Click **"Mark Complete"**
3. Enter:
  - **Completed Date:** When it was done
  - **Cost (UGX):** How much it cost
  - **Notes:** What was done, parts replaced, etc.
4. Click **"Save"**

## Maintenance Alerts

- System generates an alert when maintenance is due
- Default: 7 days before scheduled date
- Configurable in [Settings](#) → [Alerts](#)

## Alerts & Notifications

### Alert Types

| Alert              | Description                       | Default Trigger |
|--------------------|-----------------------------------|-----------------|
| Low Battery        | Battery below threshold           | < 20%           |
| Offline            | No GPS signal received            | > 30 minutes    |
| Maintenance Due    | Scheduled maintenance approaching | 7 days before   |
| Geofence Exit      | Vehicle left designated area      | On exit         |
| High Temperature   | Battery overheating               | > 45°C          |
| Rapid Battery Drop | Unexpected battery drain          | > 10% in 15 min |

### Acknowledging Alerts

1. Click the **alert** in the panel or alerts page
2. Click **"Acknowledge"**
3. The alert moves to "Acknowledged" status
4. You can add a note explaining action taken

### Configuring Alert Thresholds

1. Go to [Settings](#) → [Alerts](#)
2. For each alert type, adjust:
  - **Threshold** (e.g., change low battery from 20% to 25%)



- **Email notifications** (on/off)
- **SMS notifications** (on/off and enter phone number)
- **In-app notifications** (on/off)

3. Click "**Save**"

## SMS Notifications

- SMS uses Twilio (requires balance)
  - Standard SMS rates apply (~UGX 150/SMS)
  - Configure your phone number in **Settings** → **Organization**
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## Reports

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### Fleet Summary Report

Shows fleet-wide metrics for a date range:

- Total distance covered (km)
- Total trips
- Average fleet utilization (%)
- Total energy consumed (kwh)
- Total charging sessions
- Cost savings vs diesel

### Cost Savings Report

Calculates how much money you've saved by using EVs:

- **Fuel saved** = (Distance covered ÷ Average diesel km/liter)
- **Cost saved** = Fuel saved × Diesel price per liter (default: UGX 5,500)
- **CO<sub>2</sub> saved** = Fuel saved × Emission factor (default: 2.68 kg/liter)

### Vehicle Utilization Report

Per vehicle:

- Total distance (km)
- Number of trips
- Utilization % (time in transit vs idle)
- Energy consumed (kWh)
- Cost savings

### Exporting Reports

1. Select the report type
  2. Set date range
  3. Click "**Export PDF**" (formatted document) or "**Export CSV**" (raw data)
  4. The file downloads to your device
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## Settings

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### Organization Settings

- **Name:** Your company/organization name
- **Address:** Physical address
- **Phone:** Contact number
- **Logo:** Upload your company logo
- **Default Diesel Price:** Used for cost savings calculations
- **Default Emission Factor:** Used for CO<sub>2</sub> calculations
- **Timezone:** Set to "Africa/Kampala" for Uganda

### User Management

- See all users in your organization
- Invite new users
- Change user roles
- Deactivate users

### Alert Configuration

- Set thresholds for each alert type
- Configure notification channels (email/in-app/SMS)

### Telemetry API Keys

- Generate API keys for GPS devices
  - Each key is linked to your organization
  - Devices use this key to send telemetry data
  - If compromised, revoke and generate a new key
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## Roles & Permissions

| Action             | Admin                    | Fleet Manager            | Driver                   | Viewer                   |
|--------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| View dashboard     | <input type="checkbox"/> | <input type="checkbox"/> | Own vehicle only         | <input type="checkbox"/> |
| Add/edit vehicles  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Delete vehicles    | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Add/edit drivers   | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Assign drivers     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| View battery data  | <input type="checkbox"/> | <input type="checkbox"/> | Own vehicle only         | <input type="checkbox"/> |
| Acknowledge alerts | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Configure alerts   | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| View reports       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Export data        | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Manage users       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Manage billing     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## GPS Device Setup

### Supported Devices

- **Teltonika FMB920** (recommended for vehicles)
- **Teltonika FMB120** (with CAN bus support)
- **Generic OBD-II** telematics devices that support HTTP POST
- **Smartphone app** (for manual location updates)

### Configuration Steps

1. Generate a **Telemetry API Key** in FleetSpark (Settings → Telemetry Keys)
2. Assign the device to a vehicle (Vehicle Settings → Device ID)
3. Configure the GPS device to send HTTP POST to:  
POST `https://fleetspark.syncspherellc.com/api/v1/telemetry` Headers: X-API-Key: <your\_api\_key>  
Body: { "deviceId": "...", "timestamp": "...", "gps": {...}, "battery": {...} }
4. Set telemetry interval to **30 seconds**
5. Verify: The vehicle should appear on the map within 1-2 minutes

### Demo / Testing Mode

If you don't have GPS hardware yet:

1. Go to **Settings → Demo Mode**
2. Click **"Generate Demo Data"**

3. System creates 5 simulated vehicles with fake GPS movement
4. Useful for testing the dashboard before hardware arrives

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## FAQ

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**Q: How often does the location update?**

A: Every 30 seconds (configurable on the GPS device).

**Q: What happens if internet is down?**

A: The GPS device buffers data and sends reconnection. FleetSpark stores last known location and shows "Offline" status.

**Q: Can I track both EVs and diesel vehicles?**

A: Yes! FleetSpark works with any vehicle type. For non-EVs, battery features are simply not shown. For EVs, you also get charging and cost savings.

**Q: Can my drivers use their phones instead of GPS hardware?**

A: Yes. The mobile web app allows drivers to manually update location. Accuracy depends on phone GPS.

**Q: How is cost savings calculated?**

A:  $(\text{Distance covered in km}) \div (\text{your diesel vehicle's km/liter}) \times (\text{diesel price per liter})$ . Set your diesel price in Settings → Organization.

**Q: Can I see historical data from before I started using FleetSpark?**

A: No. FleetSpark only stores data from the moment you start sending telemetry. For historical data, export from your old system and import via CSV (v2 feature).

**Q: How many vehicles can I track?**

A: Up to 200 per organization on the Starter plan. Enterprise plans support unlimited vehicles.

**Q: Is my data secure?**

A: Yes. All data is encrypted in transit (HTTPS) and at rest. Each organization's data is isolated. We never share your data.

**Q: Can I white-label FleetSpark with my own branding?**

A: Yes, on the Enterprise plan. Contact us for pricing.

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## Support

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- **Email:** support@fleetspark.syncsphereinc.com
- **Phone:** +1 (432) 692-0996
- **Hours:** Monday – Friday, 9AM – 6PM EAT
- **Emergency:** For critical issues (system down), call the emergency line

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*FleetSpark v1.0 — Built by SyncSphere LLC*